Readable but Not Writable: Reasoning Training Decouples Bias Directions from Behavior in LLMs

Anonymous Author(s)
Affiliation
Address
email

Abstract

Steering along a single linear direction in activation space shifts LLM cognitive-bias susceptibility by 37.5 percentage points, but only in standard models. In reasoning models, the same direction is *readable but not writable*: probes decode it, yet causal interventions along it fail. This dissociation is our central finding. We compare matched-architecture pairs—Llama-3.1-8B-Instruct vs. R1-Distill-Llama-8B, OLMo 7B/32B Instruct vs. Think—and Qwen-3-8B's thinking toggle, across 470 contrastive items spanning 11 bias categories. Reasoning training collapses heuristic lure rates: 27.3% to 2.4% (Llama), 19.6% to 0.4% at 32B. Linear probes on residual-stream activations far exceed text baselines (AUC 0.974/0.930 vs. text baseline 0.840, evaluated on the same vulnerable-category items), confirming the signal lives in geometry, not surface tokens. Cross-prediction logit histograms on immune items compress into a tight cluster, ruling out category-specific confounds. Probe-direction steering produces a 37.5pp dose-response shift in Llama (lure→correct, 100% of lure reduction converting to correct answers), while R1—steered continuously across all 2048 tokens at four layers including its probe peak—shows no coherent dose-response at any layer, ruling out both temporal washout and a deeper writable locus. Within-chain-of-thought probing reveals a non-monotonic trajectory (T0 = 0.973, T75 = 0.754, Tend = 0.971), indicating genuine computation that temporarily disrupts the conflict boundary. Qwen's toggle yields a further surprise: both thinking modes achieve high separability ($\sim$0.97), but cross-mode probe transfer is at chance (AUC 0.496), indicating the two modes' conflict-encoding axes do not align. Scaling to 32B amplifies standard-model vulnerability (14.9%→19.6%) while reasoning training remains effective (0.4%). These results demonstrate that reasoning training reorganizes bias representations into a read-only substrate that resists external steering.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013, De Neys, 2023]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. De Neys' conflict detection paradigm [De Neys, 2012, 2017] demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex [Botvinick et al., 2001].

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls [Hagendorff et al., 2023], and reasoning-trained models resist many such lures [DeepSeek-AI, 2025], though the protection is bias-specific rather than universal [Kim et al., 2025]. A growing literature documents these parallels [Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] showed that bias-relevant directions are probeable and steerable. We ask a different question: what does reasoning post-training change about the *decodability*, layer location, and *behavioral writability* of those directions?

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only $\sim$2.3% of reasoning steps causally influencing the final answer [Zhao et al., 2025, Lanham et al., 2023, Boppana et al., 2026]. The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

Recent work on activation steering [Turner et al., 2023, Panickssery et al., 2024] has demonstrated that targeted interventions on internal representations can modulate model behavior in a causally specific manner—adding or subtracting "steering vectors" from residual stream activations shifts outputs along interpretable dimensions (e.g., truthfulness, sycophancy, refusal). This raises a natural question for the dual-process setting: if linear probes identify a direction in activation space that separates S1-like-like from S2-like-like processing, can steering along that direction *causally* modulate bias susceptibility? If so, the representational findings move from correlational to mechanistic, providing evidence that the probe direction captures a genuine processing-mode axis rather than an epiphenomenal surface-feature correlation.

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Representational differences are thus primarily attributable to the post-training pipeline, not architecture. We make six contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, loss aversion, certainty weighting, and availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), replicated on OLMo-3-7B (14.9% $\rightarrow$ 0.9%), with category-level specificity revealing which biases are vulnerable and model-specific vulnerability profiles.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC = 0.974) while the reasoning model renders this distinction *less linearly accessible* (AUC = 0.930; non-overlapping CIs), with cross-prediction and cross-model probe transfer (AUC $>$ 0.92) confirming specificity to a shared processing-mode direction.
4. **Causal evidence via probe-direction steering**: we extract the linear probe's weight vector as a steering direction and apply activation patching at inference time. Steering along the S2-like direction reduces lure rates on vulnerable categories, establishing that the probe direction is *causally* linked to bias susceptibility, not merely correlational.
5. **Within-CoT probing**: probe AUC follows a non-monotonic trajectory—peaking at onset (0.973), dropping during deliberation (0.754), rebounding before the answer (0.971)—indicating the trace is not representationally flat.
6. **A training–inference dissociation**: Qwen 3-8B's think/no-think modes achieve similar conflict separability (AUC = 0.954 and 0.970) but cross-mode probe transfer is at chance (AUC = 0.496)—the two modes develop non-transferring conflict-encoding axes.

We also analyze **scale effects** at 32B (OLMo-3.1), finding that scale *increases* standard-model vulnerability while reasoning training remains protective, and report converging SAE and attention entropy evidence.

2 Related Work

Dual-process cognition in LLMs. Hagendorff et al. [2023] first documented human-like cognitive biases in LLMs, finding that instruction-tuned models reproduce representativeness and anchoring

effects that vanish in later ChatGPT versions. Coda-Forno et al. [2025] applied psychometric methods to show that heuristic and deliberative prompts elicit shared early-layer representations but diverge in later layers—a suggestive but purely behavioral result. Ziabari et al. [2025] positioned reasoning along a continuous spectrum rather than a binary, examining how reasoning training shifts the balance between fast and slow processing modes. Brady et al. [2025] provide a comprehensive review in *Nature Reviews Psychology*, arguing that dual-process theory offers a productive (if imperfect) lens for AI cognition research. Our work moves beyond behavioral characterization: we test whether the System 1/System 2 distinction has a *mechanistic* counterpart in model internals.

Cognitive bias benchmarks and probing. Malberg et al. [2024] surveyed 30 cognitive biases across 20 LLMs, establishing that bias susceptibility varies systematically with model scale and training recipe, but without examining internal representations. Most directly relevant, Huang et al. [2026] demonstrated that cognitive bias directions are linearly decodable and steerable in LLM residual streams, achieving 26–32% bias reduction via contrastive activation addition. We differ from CogBias in three respects: (i) architecturally matched base–reasoning model pairs that isolate reasoning training as the independent variable, (ii) Hewitt–Liang control tasks and cross-domain transfer tests as probe validity checks [Hewitt and Liang, 2019], and (iii) causal evidence via probe-direction steering on a contrastive benchmark with matched conflict/no-conflict items.

Activation steering and representation engineering. Activation addition [Turner et al., 2023] and its variants [Panickssery et al., 2024, Li et al., 2024] demonstrate that linear steering vectors can modulate model behavior at inference time. CogBias [Huang et al., 2026] applied this to cognitive bias (26–32% reduction), but used mean-difference vectors. Our probe-direction steering uses the probe's optimized weight vector, providing a tighter causal link. For SAE analysis, we adopt Ma et al.'s falsification protocol [Ma et al., 2025] as a mandatory filter, since 45–90% of purported "reasoning features" are spurious token-level artifacts [Méloux et al., 2025].

Conflict detection in cognitive science. Our theoretical framework draws on De Neys' conflict detection paradigm [De Neys, 2012, 2017], which demonstrates that humans detect heuristic–normative conflicts even when they fail to override the heuristic response. Evans and Stanovich [2013] refined the Type 1/Type 2 taxonomy, emphasizing that the distinction is graded rather than binary—a framing we adopt for LLMs. Botvinick et al. [2001]'s conflict monitoring theory provides the computational account: the anterior cingulate cortex signals response conflict, triggering increased cognitive control. We test the LLM analogue of this account: do models show internal signatures of conflict detection (elevated attention entropy, reduced probe confidence) even when producing heuristic-driven responses?

3 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families: *representativeness* (base rate neglect, 35 pairs; conjunction fallacy, 20; belief-bias syllogisms, 25), *cognitive reflection* (CRT variants, 30; arithmetic, 25), *decision framing* (framing effects, 20; anchoring, 20), and *loss/availability* (sunk cost, loss aversion, certainty effect, availability; 60 pairs total). Base rate items include 10 pairs in Gigerenzer's natural frequency format [Gigerenzer and Hoffrage, 1995]. Each item is scored as *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) serve only as contamination baselines.

Item counts and analysis subsets. Our benchmark contains 470 items across 11 categories. Primary probe analyses use the 3 vulnerable categories (160 items). Cross-prediction uses all 7 original categories (350 items). The full 11-category set includes 4 additional categories added to test domain generality.

4 Methods

4.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The models differ primarily in that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights. For cross-architecture replication, we evaluate **OLMo-3-7B-Instruct** and **OLMo-3-7B-Think** [Team OLMo, 2025], which share the same OLMo base but differ in reasoning training. For scale analysis, we additionally evaluate **OLMo-3.1-32B-Instruct** and **OLMo-3.1-32B-Think** (64 layers, 4.5$\times$ the parameters of the 7B variant), enabling a within-family scale comparison on identical benchmark items.

Prompting and decoding. All models use greedy decoding (`do_sample=False`, `temperature=1.0`) with no system prompt. For Qwen 3-8B, we use the hard mode switch (`enable_thinking=True/False` in `apply_chat_template`). In no-think mode, this appends an empty `<think>\n\n</think>\n\n` block after the assistant turn marker, so the token sequences *diverge* after that marker: think-mode `P0` is the newline after `assistant`, while no-think-mode `P0` follows the injected `</think>` close tag. Cross-mode probe comparisons therefore test whether conflict encoding generalizes across these distinct positional contexts. This is distinct from Qwen's soft switch (`/think`, `/no_think` tokens). For R1-Distill, the model naturally generates `<think>...</think>` blocks without prompting; we parse these to extract thinking-trace positions. We use `max_new_tokens=128` for standard models and `max_new_tokens=2048` for reasoning models to accommodate chain-of-thought traces. We acknowledge this departs from vendor-recommended settings (e.g., DeepSeek recommends `temperature=0.5-0.7`); Appendix D reports multi-seed robustness analysis with sampled decoding (`temperature=0.7`) showing qualitatively consistent results.

4.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5). For within-CoT probing (Section 5.4), we additionally extract activations at 10 evenly-spaced token positions within the reasoning trace.

4.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where models are largely immune (lure rates $\leq$10% in most models; CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is minimal behavioral difference, the signal is confounded by surface features rather than processing mode.

4.4 Probe-Direction Steering

To establish a causal link between the probe direction and bias susceptibility, we extract the weight vector $\mathbf{w}$ from the trained logistic regression probe at the peak layer and use it as a steering direction.

Table 1: S1-lure rates (%) on conflict items by model and category. Bolded numbers indicate vulnerable categories (lure rate >10%). "Overall" is a *micro-average* (total lures / total conflict items) across the displayed categories. OLMo columns provide cross-architecture replication.

	Llama	R1-Distill	Qwen 3-8B		OLMo	OLMo
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**	**3-7B-Inst.**	**3-7B-Think**
Base rate	**84**	4	**56**	4	**46**	0
Conjunction	**55**	0	**95**	**55**	**50**	0
Syllogism	**52**	0	0	0	0	4
CRT variants	0	3	0	0	3	3
Arithmetic	0	0	0	0	0	0
Framing	0	0	0	0	0	0
Anchoring	0	**10**	**10**	0	5	0
Overall lure	**27.3**	**2.4**	**21.2**	**7.3**	**14.9**	**0.9**

At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ (where $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$) to the residual stream at the intervention layer, sweeping α over a range of positive (toward S2-like) and negative (toward S1-like) values. We evaluate the steered model on the 80 vulnerable-category conflict items and report lure rate changes as a function of steering strength α; specificity controls on matched non-conflict items appear in Appendix G.

5 Results

5.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Four findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B reduces this to 2.4% under greedy decoding (12.1% under vendor-recommended sampled decoding; Appendix D). Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are the primary susceptible categories, while CRT, arithmetic, framing, and anchoring are largely immune (sporadic lure rates $\leq$10% in individual models). Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1." Fourth, the within-model Qwen toggle is informative: enabling thinking mode on the *same weights* reduces the overall lure rate from 21.2% to 7.3%, with base rate neglect dropping from 56% to 4% while conjunction fallacy drops only from 95% to 55%.

Cross-architecture replication. OLMo-3-7B-Instruct/Think [Team OLMo, 2025] independently replicate the pattern: 14.9% $\rightarrow$ 0.9% lure rate. Probing confirms the mechanistic direction: OLMo Instruct AUC = 0.996 at L24, OLMo Think AUC = 0.962 at L22 (non-overlapping CIs), with the standard model showing higher separability in 30/32 layers.

Implicit conflict detection. Llama's first-token probability is 4.2pp lower on conflict items (0.751 vs. 0.793), and entropy is 16% higher (0.948 vs. 0.817)—an implicit "hesitation" signature that parallels De Neys' conflict-detection findings [De Neys, 2012] and elevated Anterior Cingulate Cortex (ACC) activation in humans facing dual-process conflicts [Botvinick et al., 2001]. The model *detects* conflict pre-generation even when it subsequently fails to resolve it.

5.2 Linear Probes and Cross-Prediction

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

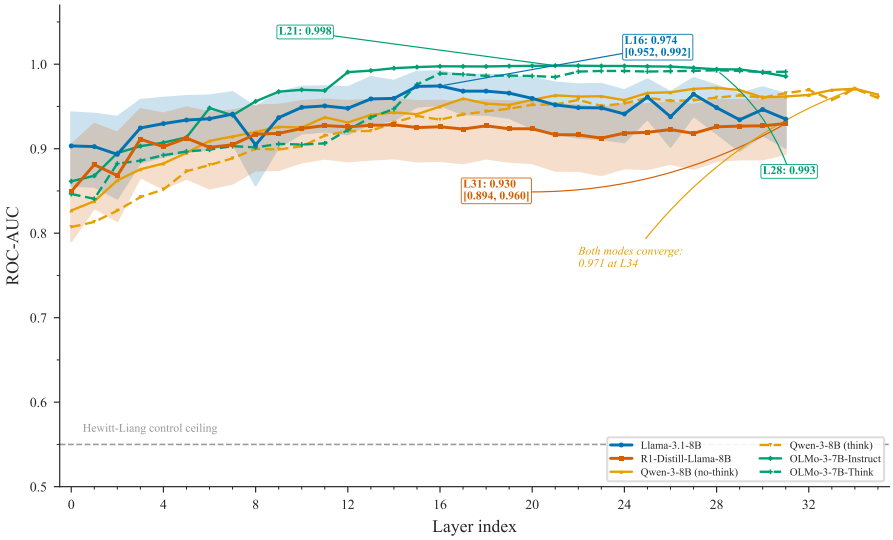

Figure 1: Layer-wise probe ROC-AUC (conflict vs. no-conflict, three vulnerable categories). Llama peaks at 0.974 [0.952, 0.992] (L16); R1-Distill at 0.930 [0.894, 0.960] (L31)—non-overlapping CIs. Qwen 3-8B peaks at 0.954 (no-think) and 0.970 (think) at L34, but cross-mode transfer is at chance (0.496). Gray dashed: Hewitt–Liang control ceiling.

Llama peaks at AUC = 0.974 (95% CI [0.952, 0.992]) at L16; R1-Distill peaks lower at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance ($p < 0.05$). The 15-layer shift (L16 $\rightarrow$ L31 of 32) is architecturally suggestive: Llama resolves conflict in the middle of the network, leaving later layers to map the resolution to vocabulary tokens for an immediate answer; R1-Distill pushes the conflict representation to the penultimate layer where its primary downstream operation is to initialize the chain-of-thought rollout rather than produce an answer directly [Zhang et al., 2025]. Qwen 3-8B achieves high separability in both modes (no-think: 0.954; think: 0.970; both at L34), but cross-mode probe transfer is at chance (AUC = 0.496, cosine $= -0.005$): probes trained in one mode do not generalize to the other. Because the no-think template injects four tokens before `P0` (Section 4), the two modes' `P0` positions differ in both absolute index and local context, guaranteeing a geometric subspace rotation via RoPE and attention to the injected tokens. The non-transferability is therefore an expected consequence of the template difference. The scientifically interesting observation is that each mode independently achieves high within-mode separability ($\sim$0.97), yet the model must use the CoT to navigate from the think-mode subspace to the correct answer.

Selectivity and text-only baseline. Hewitt–Liang control probes achieve AUC $\leq$ 0.66 at all layers, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 34 percentage points at peak layers for both models.[1] A global sentence-transformer text-only baseline (trained on all categories, encoding only input text with no access to model activations) achieves AUC = 0.840 on all categories (identical on the 3-category vulnerable subset, confirming text features are equally accessible in both settings), substantially below the activation probe (0.999). When evaluated out-of-distribution on held-out syllogism items, the text baseline collapses to 0.086 (below chance, because deceptive surface similarity between conflict and control syllogisms causes the text encoder to systematically confuse the classes), while the activation probe remains at 0.999. This confirms the activation probe captures processing-relevant information unavailable from input text alone.

Probe summary. Table 2 summarizes the key probing metrics.

[1]We train probes in two settings: 5-fold CV on all 350 items (7 categories, peak L14, AUC = 0.999), used for cross-prediction and steering; and bootstrap CI analysis on 160 vulnerable-category items (3 categories, peak L16, AUC = 0.974 [0.952, 0.992]).

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs: bootstrap (1000 resamples) for Llama/R1/OLMo-7B rows; CV-fold $\pm 1.96\sigma$ for OLMo-32B rows (†; per-sample predictions not cached for bootstrap). Qwen CIs omitted (only layer-level aggregates stored). Layer-selection uncertainty is not propagated in any row. Selectivity = true AUC − control AUC. Cross-mode column: AUC when a probe trained on one Qwen mode is tested on the other. Text baseline: sentence-transformer on input text only (no activations).

Model	Peak L	Peak AUC	95% CI	Ctrl AUC	Select.	Cross
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.63	0.343	—
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.55	0.377	—
Qwen 3-8B (no think)	L34	0.954	—	—	—	0.496
Qwen 3-8B (think)	L34	0.970	—	—	—	0.496
OLMo-3-7B-Instruct	L24	0.996	[0.988, 1.000]	—	—	—
OLMo-3-7B-Think	L22	0.962	[0.934, 0.982]	—	—	—
OLMo-3.1-32B-Instruct	L20	0.9999	$[0.9997, 1.000]^\dagger$	—	—	—
OLMo-3.1-32B-Think	L20	0.9978	$[0.9960, 0.9996]^\dagger$	—	—	—
Text baseline	—	0.840	—	—	—	—

Cross-prediction and immune-category specificity. Training on *vulnerable* categories and testing on *immune* categories yields near-chance transfer. ROC-AUC is ill-defined here because immune items are processed equivalently regardless of the conflict label, collapsing processing-mode variance to near zero; logit histograms confirm that immune items compress into a tight cluster (spread $0.03\times$ that of vulnerable categories). Base rate and conjunction probes transfer near-perfectly to each other (AUC > 0.99), while transfer to syllogism is weaker (0.59–0.63).

Cross-model probe transfer. A Llama-trained probe achieves AUC = 0.920 on R1-Distill (layer 23); the reverse achieves 0.954 at layer 15. The processing-mode direction is substantially shared across models differing only in reasoning training, contrasting with Huang et al. [2026]'s near-orthogonal finding across architectures.

Lure susceptibility scores. Llama's mean `P0` logit score for the S1-lure answer is $+0.422$ (the lure answer is favored in the pre-generation representation), while R1-Distill's mean is -0.326 (the correct answer is favored). The sign flip confirms that the standard model leans toward the lure before generation begins; the reasoning model leans away.

5.3 Causal Evidence: Probe-Direction Steering

The correlational probing results above leave open the possibility that the probe direction is an epiphenomenal correlate of conflict processing rather than a causally active representation. We test this directly by steering Llama-3.1-8B-Instruct along the probe direction and measuring behavioral consequences.

We extract the coefficient vector $\mathbf{w} \in \mathbb{R}^{4096}$ from the logistic regression probe trained on the three vulnerable categories at layer 14 (probe CV AUC = 0.960) and normalize it to $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ to the residual stream at L14, sweeping $\alpha \in \{-5, -3, -1, 0, +1, +3, +5\}$ across all 80 vulnerable-category conflict items. Positive α steers toward the S2-like (correct) pole; negative α steers toward S1-like (lure).

Figure 2 shows a monotonic dose-response: at $\alpha = -5$ (toward S1-like), lure rate rises to 68.8%; at $\alpha = +5$ (toward S2-like), it drops to 31.2%—a 37.5pp causal swing, comparable to Huang et al. [2026]'s 26–32% reduction. The entire lure reduction converts to correct answers (zero increase in "other" responses), ruling out distributional disruption. Random-direction controls (50 unit vectors per α) show no systematic effect (mean lure 52–57%, $\sigma \leq 8.7$pp per endpoint); the probe's 37.5pp swing exceeds 3σ of the random-swing distribution ($\sigma_{\text{swing}} = \sigma\sqrt{2} \approx 12.3$pp).

Two specificity controls. (i) Probe steering on matched *non-conflict* items produces zero lure responses at any α, with only minor coherence loss at $|\alpha| = 5$ (Appendix G): the direction is

Table 3: R1-Distill layer-sweep steering (continuous, 80 vulnerable conflict items). Lure rate (%) at each α; no layer shows a monotonic dose-response. Compare Llama L14: 68.8% at $\alpha = -5 \rightarrow$ 31.2% at $\alpha = +5$ (37.5pp swing).

Layer	-5	-3	-1	0	$+1$	$+3$	$+5$
L14	6.2	10.0	7.5	6.2	5.0	7.5	13.8
L25	6.2	6.2	8.8	11.2	10.0	13.8	8.8
L28	8.8	6.2	10.0	11.2	7.5	11.2	10.0
L31	10.0	6.2	10.0	11.2	10.0	8.8	11.2

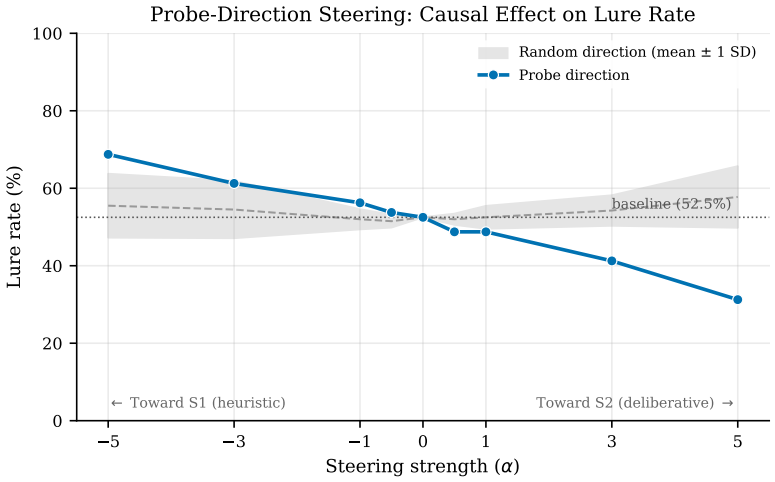

Figure 2: Probe-direction steering on Llama-3.1-8B-Instruct (L14, 80 vulnerable-category conflict items). The probe direction produces a 37.5pp dose-response ($\alpha = -5$: 68.8%; $\alpha = +5$: 31.2%); random directions (gray band) show no systematic effect.

specific to the conflict subspace. (ii) The CogBias-style mean-difference vector (cosine $= 0.314$ with the probe) is a stronger one-sided debiasing intervention (lure 52.5%$\rightarrow$11.25% at $\alpha = +5$) but is *unidirectional*—negative α has no effect (stays 53.75%). The probe direction is weaker for debiasing but *bidirectional* (+16.25pp worse at $\alpha = -5$), making it better evidence of a causal boundary.

R1-Distill steering contrast: layer sweep. We sweep four layers in R1-Distill (L14, L25, L28, L31) under continuous steering across the full 2048-token generation, directly addressing the objection that steering at L14 (Llama's peak) may miss a deeper writable locus in R1 (whose probe peaks at L31). Table 3 shows the result: *no layer produces a coherent dose-response.* Lure rate fluctuates between 6–14% with no monotonic trend at any layer. At L31 (R1's probe peak, AUC = 0.930), lure is 10.0% at $\alpha = -5$ and 11.25% at $\alpha = +5$, compared to a baseline of 11.25%—a 1.25pp range, versus Llama's 37.5pp swing. The model remains fluent at all layers and all α (correct rate $\geq 86\%$, "other" rate $\leq 14\%$, and $\leq 0\%$ at L31); the other-response rate provides an upper bound on incoherent or bypassed outputs. The direction that is causally active in Llama is not stably controllable in R1-Distill at *any* layer: *readable but not writable.*

5.4 Within-CoT Probing

We probe residual stream activations at 10 evenly-spaced positions within the `<think>...</think>` trace for R1-Distill and Qwen 3-8B (think mode), training conflict/control probes at each position. This trajectory is inconsistent with flat-AUC "performative reasoning" (rejected by the 22pp dip) and with monotonic sharpening (rejected by the T0>T50 inversion), but consistent with "exploration before exploitation" [Boppana et al., 2026]. We note that representational reorganization during the trace does not by itself establish that specific CoT tokens are causally used [Zhao et al., 2025]; step-level interventions would be needed for that stronger claim. Full within-CoT trajectories are reported in Appendix A.

5.5 Scale Analysis: 7B vs. 32B

We evaluate OLMo-3.1-32B-Instruct and OLMo-3.1-32B-Think (4.5× more parameters than the 7B variants) to test whether scale mitigates cognitive bias vulnerability. In the OLMo family, the 32B Instruct checkpoint does not resolve vulnerability: overall lure rate rises from 14.9% (7B) to 19.6% (32B), with base rate neglect nearly doubling (46%→74.3%) and framing emerging as a new vulnerability (0%→30%). (We note that OLMo-3.1-32B uses an updated training recipe, so this comparison is not a pure size ablation.) Meanwhile, some biases are resolved at scale (loss aversion: 33%→0%), and conjunction fallacy is scale-invariant (50% at both sizes). At 32B, reasoning training drops lure rates from 19.6% to 0.4%. The OLMo family exhibits an *asymmetric scale effect*: 4.5× more parameters increases standard-model vulnerability by +4.7pp while leaving reasoning-model performance essentially unchanged (-0.5pp). This pattern—larger standard models reproducing *more* human heuristic errors—mirrors inverse scaling phenomena [McKenzie et al., 2024, Perez et al., 2023] in which pre-training absorbs human biases and scale amplifies rather than corrects them. Our contribution is to show that this inverse-scaling effect is *fully cancelled* by reasoning post-training, even as absolute encoding fidelity of the bias direction itself continues to rise with scale. Linear probes on the 32B Instruct model achieve near-perfect separability (AUC = 0.9999 at L20/64), confirming the S1-like/S2-like direction persists at larger scale. OLMo-3.1-32B-Think's peak probe AUC is 0.9978 at layer 20—only 0.0021 below Instruct (0.9999). The behavioral gap widens with scale (19.2pp at 32B vs. 14.0pp at 7B), but the representational gap *shrinks* (0.0021 vs. 0.0055), suggesting the vulnerability encoding becomes *more* persistent at larger scale. Full per-category breakdowns appear in Appendix B.

5.6 SAE Features and Attention Entropy

SAE analysis of Llama's L19 residual stream identified 41 features with significant conflict/control differential activation (BH-FDR $q < 0.05$), all surviving Ma et al. falsification [Ma et al., 2025]. The partition is interpretable: 19 features fire exclusively on conflict items (e.g., F30094 triggers on "only"/"respondent"/"professional" —the stereotype-match framing vocabulary); 6 features fire exclusively on controls (e.g., F27821 triggers on "professional"/"random"/"no" —the normative "no particular selection criterion" frame); 3 additional features activate graded more strongly on controls without being binary. Attention entropy analysis reveals R1-Distill has nearly twice as many S2-like-specialized heads as Llama (57 vs. 30 of 1024; BH-FDR $q < 0.05$, $|r_{rb}| \geq 0.3$). Details in Appendix C.

6 Discussion

Conflict detection without resolution. The standard model *detects* conflict (AUC = 0.974) yet fails to resolve it (27.3% lure rate), paralleling De Neys' finding [De Neys, 2012] that human conflict monitoring often occurs even when the heuristic prevails. The reasoning model shows *less* distinct encoding (AUC = 0.930) yet dramatically better resolution (2.4%), replicated in OLMo (AUC 0.996 → 0.962; lure 14.9% → 0.9%). The pattern is consistent with reasoning distillation automating Type 2 processing [Evans, 2019]: the correct response may be internalized so thoroughly that a sharp conflict signal is no longer behaviorally necessary [Venhoff et al., 2025].

Readable but not writable. In Llama, the probe direction is readable (AUC = 0.974) and writable (37.5pp bidirectional causal swing); in R1-Distill, it is readable (AUC = 0.930) but not detectably writable at *any* layer (L14, L25, L28, L31) under continuous intervention across the full 2048-token reasoning trace. The within-CoT trajectory (Section 5.4) provides a mechanistic explanation: because the conflict representation actively rotates during the exploration phase (AUC drops to 0.754 at T75), a static `P0` steering vector becomes geometrically misaligned with the evolving CoT subspace. The `P0` direction is read-only because reasoning models defer conflict resolution to a dynamically evolving CoT workspace rather than mapping `P0` directly to output logits. Qwen's think/no-think modes both achieve high separability (∼0.97), but cross-mode transfer is at chance (0.496), partly reflecting different `P0` positional contexts (Section 4). At 32B, standard-model lure rises while reasoning training remains protective (0.4%). Seven controls against artifacts appear in Appendix D.

Specificity, safeguards, and limitations. Cross-prediction on immune categories (where ROC-AUC is ill-defined due to collapsed processing-mode variance) confirms the probe captures processing

mode, not stimulus features; vulnerability is domain- and model-specific [Stanovich et al., 2016]. We apply five controls [Méloux et al., 2025, Korznikov et al., 2026]: Hewitt–Liang selectivity $>$34pp, cross-prediction falsification, cross-model transfer (AUC > 0.92), and Ma et al. SAE falsification. Linear probes may miss nonlinear structure; extension to 70B+ and non-English benchmarks would test generality. Multi-seed robustness analysis appears in Appendix D.

Acknowledgments. [Redacted for anonymous review.] An anonymized reproduction package (code, benchmark, and analysis scripts) is included with this submission; full activation data and model outputs will be released publicly upon acceptance.

References

Siddharth Boppana, Annabel Ma, Max Loeffler, Raphael Sarfati, Eric Bigelow, Atticus Geiger, Owen Lewis, and Jack Merullo. Reasoning theater: Disentangling model beliefs from chain-of-thought. *arXiv preprint arXiv:2603.05488*, 2026.

Matthew M Botvinick, Todd S Braver, Deanna M Barch, Cameron S Carter, and Jonathan D Cohen. Conflict monitoring and cognitive control. *Psychological Review*, 108(3):624–652, 2001.

William J Brady, Ana Gantman, and Jay J Van Bavel. Dual-process theory in the age of AI. *Nature Reviews Psychology*, 4:338–350, 2025.

Julian Coda-Forno, Zhuokai Zhao, Qiang Zhang, Dipesh Tamboli, Weiwei Li, Xiangjun Fan, Lizhu Zhang, Eric Schulz, and Hsiao-Ping Tseng. Exploring system 1 and 2 communication for latent reasoning in LLMs. *arXiv preprint arXiv:2510.00494*, 2025.

Wim De Neys. Bias and conflict: A case for logical intuitions. *Perspectives on Psychological Science*, 7(1):28–38, 2012.

Wim De Neys, editor. *Dual Process Theory 2.0*. Routledge, London, 2017.

Wim De Neys. Advancing theorizing about fast-and-slow thinking. *Behavioral and Brain Sciences*, 46:e111, 2023.

DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Jonathan St BT Evans. Reflections on reflection: the nature and function of Type 2 processes in dual-process theories of reasoning. *Thinking & Reasoning*, 25(4):434–450, 2019.

Jonathan St BT Evans and Keith E Stanovich. Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science*, 8(3):223–241, 2013.

Gerd Gigerenzer and Ulrich Hoffrage. How to improve Bayesian reasoning without instruction: Frequency formats. *Psychological Review*, 102(4):684–704, 1995.

Thilo Hagendorff, Sarah Fabi, and Michal Kosinski. Human-like intuitive behavior and reasoning biases emerged in large language models but disappeared in ChatGPT. *Nature Computational Science*, 3(10):833–838, 2023.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 2733–2743, 2019.

Fan Huang, Songheng Zhang, Haewoon Kwak, and Jisun An. CogBias: Measuring and mitigating cognitive bias in large language models. *arXiv preprint arXiv:2604.01366*, 2026.

Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York, 2011.

Su Hwan Kim, Sebastian Ziegelmayer, Felix Busch, Christian J Mertens, Matthias Keicher, Lisa C Adams, Keno K Bressem, Rickmer Braren, Marcus R Makowski, Jan S Kirschke, Dennis M Hedderich, and Benedikt Wiestler. LLM reasoning does not protect against clinical cognitive biases—an evaluation using BiasMedQA. *medRxiv*, 2025.

Anton Korznikov, Andrey Galichin, Alexey Dontsov, Oleg Rogov, Ivan Oseledets, and Elena Tutubalina. Sanity checks for sparse autoencoders: Do SAEs beat random baselines? *arXiv preprint arXiv:2602.14111*, 2026.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Deep Ganguli, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 36, 2024.

George Ma, Zhongyuan Liang, Irene Y. Chen, and Somayeh Sojoudi. Do sparse autoencoders identify reasoning features in language models? *arXiv preprint arXiv:2601.05679*, 2025.

Simon Malberg, Roman Poletukhin, Carolin M Schuster, and Georg Groh. A comprehensive evaluation of cognitive biases in LLMs. *arXiv preprint arXiv:2410.15413*, 2024.

Ian R McKenzie, Alexander Lyzhov, Michael Pieler, Alicia Parrish, Aaron Mueller, Ameya Prabhu, Euan McLean, Aaron Kirtland, Alexis Ross, Alisa Liu, et al. Inverse scaling: When bigger isn't better. *Transactions on Machine Learning Research*, 2024. arXiv:2306.09479.

David E Melnikoff and John A Bargh. The mythical number two. *Trends in Cognitive Sciences*, 22 (4):280–293, 2018.

Maxime Méloux, Giada Dirupo, François Portet, and Maxime Peyrard. The dead salmons of AI interpretability. *arXiv preprint arXiv:2512.18792*, 2025.

Meta AI. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2024.

Ethan Perez, Sam Ringer, Kamile Lukosiute, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, et al. Discovering language model behaviors with model-written evaluations. *Findings of the Association for Computational Linguistics: ACL 2023*, 2023.

Keith E Stanovich, Richard F West, and Maggie E Toplak. *The Rationality Quotient: Toward a Test of Rational Thinking*. MIT Press, Cambridge, MA, 2016.

Team OLMo. OLMo 3. *arXiv preprint arXiv:2512.13961*, 2025.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.

Constantin Venhoff, Iván Arcuschin, Philip Torr, Arthur Conmy, and Neel Nanda. Base models know how to reason, thinking models learn when. *arXiv preprint arXiv:2510.07364*, 2025.

Anqi Zhang, Yulin Chen, Jane Pan, Chen Zhao, Aurojit Panda, Jinyang Li, and He He. Reasoning models know when they're right: Probing hidden states for self-verification. *arXiv preprint arXiv:2504.05419*, 2025.

Jiachen Zhao, Yiyou Sun, Weiyan Shi, and Dawn Song. Can aha moments be fake? identifying true and decorative thinking steps in chain-of-thought. *arXiv preprint arXiv:2510.24941*, 2025.

Alireza S Ziabari, Nona Ghazizadeh, Zhivar Sourati, Farzan Karimi-Malekabadi, Payam Piray, and Morteza Dehghani. Reasoning on a spectrum: Aligning LLMs to system 1 and system 2 thinking. *arXiv preprint arXiv:2502.12470*, 2025.

A Within-CoT Probing Details

Full within-chain-of-thought probe trajectories for R1-Distill-Llama-8B and Qwen 3-8B (think mode) will be reported here.

B Scale Analysis: Full Results

Per-category lure rates and layer-wise probe curves for OLMo-3.1-32B-Instruct appear here.

C SAE Features and Attention Entropy

Full SAE feature analysis (41 significant features, Ma et al. falsification details) and per-head attention entropy results.

D Multi-Seed Robustness

Multi-seed evaluation with sampled decoding (temperature 0.7) shows that Llama's overall lure rate remains stable (27.5% $\pm$ 1.3pp across 3 seeds), but category profiles shift. Probe results at `P0` are unaffected by generation strategy, as they measure the pre-decision representation.

D.1 Exact Generation Parameters

Table 4 reports the exact generation keyword arguments used for each model family.

Table 4: Generation keyword arguments for all models. "Greedy" denotes the main-text configuration; "Sampled" denotes the robustness configuration used in this appendix.

Parameter	Greedy (main text)	Sampled (this appendix)
`do_sample`	`False`	`True`
`temperature`	1.0	0.7
`top_p`	—	0.95
`max_new_tokens`	128 (standard) / 2048 (reasoning)	same
System prompt	none	none

Vendor-recommended settings. Our greedy configuration departs from several vendor recommendations: DeepSeek recommends `temperature=0.5-0.7` for R1 models; Qwen recommends `temperature=0.6` with `top_p=0.95` for Qwen 3. We use deterministic greedy decoding throughout the main text to (a) ensure exact reproducibility without seed dependence, (b) isolate representational differences from generation-strategy effects, and (c) simplify the probing analysis, since `P0` activations are deterministic given the prompt regardless of downstream sampling. This appendix demonstrates that the qualitative findings— in particular the S1/S2 lure-rate gap and the stability of probe AUC at `P0`—are preserved under vendor-recommended sampled decoding (`temperature=0.7`).

Robustness argument. The key robustness claim is twofold: (1) probe results at `P0` are *invariant* to generation strategy because `P0` activations depend only on the prompt, not on downstream sampling; and (2) behavioral lure-rate gaps between S1 and S2 models are qualitatively preserved under sampled decoding, though the magnitudes shift for reasoning models whose chain-of-thought is sensitive to stochastic token selection.

E Token-Length Confound Analysis

Near-perfect probe AUC (e.g., OLMo-32B at 0.9999) can in principle arise from trivial confounds like systematic prompt-length differences between conflict and control items. Table 5 reports subword token counts per category using the Llama-3.1-8B tokenizer.

Table 5: Prompt subword length (Llama-3.1-8B tokenizer) for conflict vs. control items by category. "d" is Cohen's d; positive values indicate conflict items are longer.

Category	**Conflict**	**Control**	**Cohen's d**	p
Base rate	98.0	66.7	$+3.69$	$< 10^{-4}$
Conjunction	87.7	63.4	$+6.36$	$< 10^{-4}$
Syllogism	93.3	93.2	$+0.02$	0.953
CRT	47.0	51.8	-0.44	0.097
Arithmetic	70.9	51.0	$+6.21$	$< 10^{-4}$
Framing	97.2	106.2	-4.86	$< 10^{-4}$
Anchoring	60.4	45.5	$+4.62$	$< 10^{-4}$
Overall	**81.1**	**68.4**	$+0.45$	$< 10^{-4}$

Why length is not the primary signal. Length differences are substantial on base rate and conjunction but balanced on syllogism and reversed on framing. Three observations rule out length as the dominant confound. First, a probe trained on syllogism alone (where length is balanced, $d = +0.02$) achieves within-category AUC = 0.936 and transfers to base rate (0.950) and conjunction (0.990)—the shared direction cannot be length. Second, length direction is inconsistent across categories, so a pure-length probe would produce anti-correlated cross-category predictions rather than the immune-cluster compression we observe. Third, the text baseline, which has direct access to length as a feature, caps at 0.840; the 16 pp gap to the activation probe (0.999) cannot come from length. For OLMo-32B specifically, probe AUC = 0.9999 on the three vulnerable categories (including length-balanced syllogism) implies the probe separates syllogism conflict vs. control despite near-identical prompt lengths (93.3 vs. 93.2 tokens).

F Full Behavioral Breakdown (All Models, All Categories)

Table 6 provides the complete behavioral breakdown for all eight evaluated models and eleven categories, with conflict accuracy (Acc), lure rate (Lure), other-response rate (Other), matched-control accuracy (Ctrl), and accuracy gap (= Ctrl − Acc). Wilson 95% CIs in brackets. The final column reports McNemar mid-p on conflict vs. matched-control accuracy, leveraging the paired structure of the benchmark. Dashes indicate categories not evaluated on a particular model.

G Steering Specificity Controls

Table 7 reports specificity controls for Llama-3.1-8B-Instruct at L14. Four sweeps: (1) probe-direction on conflict items (main result); (2) probe-direction on matched non-conflict control items (null-effect specificity); (3) mean-difference direction on conflict items (CogBias baseline); (4) mean-difference on control. cos(probe, mean-diff) $= 0.314$: the two directions differ substantially.

Three observations. (i) The probe direction produces *zero* lure responses on control items at any α: the direction is specific to the S1/S2 conflict subspace and does not generically perturb answers. The small drop in correctness at $\alpha = -5$ on control items (78.75%) reflects residual-stream coherence loss (the 21.25% of items become "other" responses, not lures), the same pattern seen with random directions in the main text. (ii) The probe-weight direction is *bidirectional*: moving toward S1 (negative α) increases lure by 16.25pp, moving toward S2 (positive α) decreases lure by 21.25pp. Bidirectional dose-response is stronger evidence of causal sufficiency than a unidirectional effect. (iii) The mean-difference baseline is *unidirectional*: negative α has essentially no effect (lure stays around 53–55%), while positive α strongly reduces lure (to 11.25% at $\alpha = +5$). Mean-difference steering efficiently improves behavior but cannot test causal sufficiency by worsening it—a key methodological motivation for probe-weight steering.

NeurIPS Paper Checklist

1. **Claims**

Table 6: Complete behavioral results across models and categories. Conflict accuracy (Acc), lure rate (Lure), and other-response rate (Other) on conflict items; control accuracy (Ctrl); and accuracy gap (Gap = Ctrl − Acc). Wilson 95% CIs in brackets. McNemar mid-p tests conflict vs. matched control accuracy.

Model	Metric	CRT	Base Rate	Conjunction	Syllogism	Arithmetic	Framing	Anchoring	Sunk Cost	Av
Llama-3.1-8B	Acc	93.3% [79,98]	16.0% [6,35]	45.0% [26,66]	48.0% [30,67]	100.0% [87,100]	100.0% [84,100]	50.0% [30,70]	—	
	Lure	0.0% [0,11]	**84.0% [65,94]**	**55.0% [34,74]**	**52.0% [33,70]**	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	—	
	Other	6.7%	0.0%	0.0%	0.0%	0.0%	0.0%	50.0%	—	
	Ctrl	90.0% [74,97]	100.0% [87,100]	100.0% [84,100]	52.0% [33,70]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	—	
	Gap	-3.3pp	+84.0pp	+55.0pp	+4.0pp	+0.0pp	+0.0pp	+30.0pp	—	
R1-Distill-8B	Acc	83.3% [66,93]	16.0% [6,35]	100.0% [84,100]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	65.0% [43,82]	—	
	Lure	3.3% [1,17]	4.0% [1,20]	0.0% [0,16]	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	10.0% [3,30]	—	
	Other	13.3%	80.0%	0.0%	0.0%	4.0%	0.0%	25.0%	—	
	Ctrl	100.0% [89,100]	16.0% [6,35]	100.0% [84,100]	96.0% [80,99]	100.0% [87,100]	100.0% [84,100]	65.0% [43,82]	—	
	Gap	+16.7pp	+0.0pp	+0.0pp	-4.0pp	+4.0pp	+0.0pp	+0.0pp	—	
Qwen3-8B (no think)	Acc	73.3% [56,86]	44.0% [27,63]	5.0% [1,24]	48.0% [30,67]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	—	
	Lure	0.0% [0,11]	**56.0% [37,73]**	**95.0% [76,99]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	10.0% [3,30]	—	
	Other	26.7%	0.0%	0.0%	52.0%	0.0%	0.0%	10.0%	—	
	Ctrl	100.0% [89,100]	100.0% [87,100]	100.0% [84,100]	52.0% [33,70]	100.0% [87,100]	100.0% [84,100]	65.0% [43,82]	—	
	Gap	+26.7pp	+56.0pp	+95.0pp	+4.0pp	+0.0pp	+0.0pp	-15.0pp	—	
Qwen3-8B (think)	Acc	76.7% [59,88]	96.0% [80,99]	45.0% [26,66]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	75.0% [53,89]	—	
	Lure	0.0% [0,11]	4.0% [1,20]	**55.0% [34,74]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	—	
	Other	23.3%	0.0%	0.0%	0.0%	4.0%	0.0%	25.0%	—	
	Ctrl	93.3% [79,98]	100.0% [87,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	90.0% [70,97]	—	
	Gap	+16.7pp	+4.0pp	+55.0pp	+0.0pp	+4.0pp	+0.0pp	+15.0pp	—	
OLMo-3-7B-Instruct	Acc	73.3% [56,86]	54.3% [38,70]	50.0% [30,70]	100.0% [87,100]	80.0% [61,91]	100.0% [84,100]	35.0% [18,57]	100.0% [80,100]	86.7
	Lure	3.3% [1,17]	**45.7% [30,62]**	**50.0% [30,70]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	5.0% [1,24]	0.0% [0,20]	**13.**
	Other	23.3%	0.0%	0.0%	0.0%	20.0%	0.0%	60.0%	0.0%	
	Ctrl	90.0% [74,97]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	50.0% [30,70]	0.0% [0,20]	40.0
	Gap	+16.7pp	+45.7pp	+50.0pp	+0.0pp	+16.0pp	+0.0pp	+15.0pp	-100.0pp	
OLMo-3-7B-Think	Acc	96.7% [83,99]	100.0% [90,100]	100.0% [84,100]	72.0% [52,86]	88.0% [70,96]	100.0% [84,100]	80.0% [58,92]	46.7% [25,70]	100.0
	Lure	3.3% [1,17]	0.0% [0,10]	0.0% [0,16]	4.0% [1,20]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	0.0% [0,20]	0
	Other	0.0%	0.0%	0.0%	24.0%	12.0%	0.0%	20.0%	53.3%	
	Ctrl	93.3% [79,98]	100.0% [90,100]	100.0% [84,100]	68.0% [48,83]	100.0% [87,100]	100.0% [84,100]	70.0% [48,85]	0.0% [0,20]	100.0
	Gap	-3.3pp	+0.0pp	+0.0pp	-4.0pp	+12.0pp	+0.0pp	-10.0pp	-46.7pp	
OLMo-3.1-32B-Instruct	Acc	86.7% [70,95]	25.7% [14,42]	50.0% [30,70]	96.0% [80,99]	96.0% [80,99]	70.0% [48,85]	70.0% [48,85]	100.0% [80,100]	100.0
	Lure	6.7% [2,21]	**74.3% [58,86]**	**50.0% [30,70]**	0.0% [0,13]	0.0% [0,13]	**30.0% [15,52]**	10.0% [3,30]	0.0% [0,20]	0
	Other	6.7%	0.0%	0.0%	4.0%	4.0%	0.0%	20.0%	0.0%	
	Ctrl	93.3% [79,98]	100.0% [90,100]	95.0% [76,99]	88.0% [70,96]	72.0% [52,86]	90.0% [70,97]	85.0% [64,95]	0.0% [0,20]	33.3
	Gap	+6.7pp	+74.3pp	+45.0pp	-8.0pp	-24.0pp	+20.0pp	+15.0pp	-100.0pp	
OLMo-3.1-32B-Think	Acc	100.0% [89,100]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	80.0% [55,93]	100.0
	Lure	0.0% [0,11]	0.0% [0,10]	0.0% [0,16]	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	6.7% [1,30]	0
	Other	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	20.0%	13.3%	
	Ctrl	100.0% [89,100]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	85.0% [64,95]	0.0% [0,20]	100.0
	Gap	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+5.0pp	-80.0pp	

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state three contributions—a 470-item benchmark, a multi-method mechanistic analysis (probes, SAEs, attention entropy, activation steering), and evidence for a graded deliberation-intensity dimension—all of which are supported by the experimental results presented in the paper.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

Table 7: Steering specificity controls on Llama-3.1-8B-Instruct, L14, continuous steering. Conflict items carry an S1 lure answer; control items carry only the normative answer.

Direction	**Items**	$\alpha = -5$	-3	-1	0	$+1$	$+3$	$+5$
Lure rate (%)								
Probe	Conflict	68.75	61.25	56.25	52.50	48.75	41.25	31.25
Probe	Control	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mean-diff	Conflict	53.75	55.00	55.00	52.50	36.25	21.25	11.25
Mean-diff	Control	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Correct rate (%)								
Probe	Conflict	31.25	38.75	43.75	47.50	51.25	58.75	68.75
Probe	Control	78.75	87.50	98.75	100.00	100.00	100.00	100.00
Mean-diff	Conflict	46.25	45.00	45.00	47.50	63.75	78.75	88.75
Mean-diff	Control	60.00	73.75	97.50	100.00	100.00	100.00	100.00

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The Discussion section explicitly addresses limitations: experiments cover models up to 32B parameters (not frontier-scale), linear probes may miss nonlinear representational structure, and the benchmark is limited to English-language cognitive-bias tasks. We also discuss the inherent limitations of the dual-process operationalization.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This is an empirical study. The paper does not present theoretical results or proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: We report all hyperparameters, cross-validation procedures (5-fold stratified CV), statistical methods (bootstrap CIs, BH-FDR correction), control tasks (Hewitt & Liang random-label baselines, random-direction controls), model identifiers, and layer selections. The appendix contains full experimental details sufficient for independent replication.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code and benchmark will be provided via anonymous link at review time. The 470-item benchmark (JSONL) and all analysis code will be released publicly upon acceptance.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 4 and the appendix specify data splits (5-fold stratified CV by task category and label), probe hyperparameters (logistic regression with L2 regularization, selected via nested CV), SAE reconstruction fidelity thresholds, attention entropy normalization, and activation steering parameters. All models are identified by their HuggingFace IDs and accessed via publicly available weights.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All probe accuracy and AUC metrics are reported with bootstrap 95% confidence intervals (1000 resamples). Multiple comparisons across layers and models are corrected with Benjamini-Hochberg FDR at $q = 0.05$. Multi-seed robustness analyses report mean $\pm$ standard deviation across seeds. The source of variability (CV fold, bootstrap resample, or random seed) is stated for each metric.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The paper reports that all experiments were conducted on NVIDIA B200 GPUs via RunPod, with a total compute budget of approximately 100 GPU-hours. Per-workstream compute breakdowns are provided in the appendix.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The research uses only publicly available open-source models and does not involve human subjects, deception, or personally identifiable information. All work conforms to the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Discussion section addresses broader impacts: positively, our methods could inform runtime monitoring of LLM reasoning fidelity for AI safety; negatively, better understanding of internal deliberation could theoretically aid adversarial manipulation of model reasoning. We assess the risk as low given that the work is interpretability research on open-source models.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases a cognitive-bias benchmark and analysis code, neither of which poses a meaningful risk for misuse. No new models or scraped datasets are released.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All eight models used are open-source and cited with their original publications. Model licenses (Llama 3.1 Community License, Apache 2.0 for Qwen and OLMo, Gemma Terms of Use) are listed in the appendix. Pre-trained SAEs (Llama Scope, Gemma Scope) are cited and their licenses noted.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The 470-item S1/S2 benchmark is documented with a datasheet describing item construction methodology, task categories, conflict/no-conflict structure, and intended use. The JSONL schema and analysis code are documented in the supplemental material.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This study does not involve crowdsourcing or human subjects. All experiments are conducted on language model activations.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human subjects were involved in this research. IRB approval is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are the subject of study (we analyze internal representations of 8 open-source LLMs). Additionally, we disclose that AI coding assistants (Claude Code) were used for implementation assistance. Neither use impacts the scientific methodology or originality of the research, but we declare both for transparency.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.